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Tooth wear and obesity: investigating the impact of BMI, waist circumference, and oral health-related quality of life

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Abstract

This study evaluated the association between tooth-wear risk and obesity-related measures, and explored its impact on the Oral-Health-Related Quality of Life (OHRQoL) among Pakistani adults. This single-centred cross-sectional study examined tooth wear using the Basic Erosive Wear Examination (BEWE) in 406 participants visiting the dental clinics of a tertiary care hospital, aged 18–65 years. The Oral Health Impact Profile-14 (OHIP-14) questionnaire, available in English and a validated Urdu translation, was completed. Age, sex, weight, height, Body-Mass Index (BMI), and waist circumference (WC) were recorded. Data were analysed using. SPSS version 23.0. Higher BMI and WC were significantly associated with increased tooth wear severity ($p<0.001$). Thirty-five (22.3%) and 44 (28.8%) participants in the increased and higher-high risk BMI categories, respectively, had BEWE scores$\geq$14. Fifty-seven (32.8%) males and 19 (17.3%) females in the excessive risk group for WC had BEWE scores$\geq$14. Multinomial logistic regression showed that increasing BMI and WC were significantly associated with higher odds of being in the medium- and high-risk tooth wear groups. A slight increasing trend in OHIP-14 scores was observed with increasing wear severity, differences across risk groups were not statistically significant ($p=0.956$). However, specific functional impacts were observed; participants in higher tooth wear risk groups and those reporting frequent GERD symptoms showed significantly greater impairment in taste perception ($p<0.05$). Higher BMI and waist circumference were associated with higher BEWE tooth-wear categories among Pakistani adults. BMI and waist circumference represent general and central adiposity, respectively. However, the cross-sectional design precludes conclusions regarding causality or tooth-wear progression. Overall OHRQoL did not differ significantly across tooth-wear categories, although an exploratory association with worsening taste perception was observed.

Keywords Central obesity · BMI · OHIP-14 · BEWE · tooth wear · waist circumference

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Introduction

Tooth wear is a multifactorial and progressive loss of dental hard tissues, presenting as erosion, attrition, or abrasion, either independently or in combination[1,2]. Its prevalence is rising globally[3], particularly among the adult population, where its severity poses challenges to restorative dental care and compromises both aesthetics and function[4]. Clinically, severity is often assessed using the Basic Erosive Wear Examination (BEWE), a validated four-point index that records the most affected surface in each sextant[5]. The cumulative score stratifies patients by risk level and guides preventive and recall strategies[6]. While the BEWE was initially conceptualized with an emphasis on erosive wear, the index records tooth wear from all aetiologies, including attrition, abrasion, and erosion. It assesses the degree of surface loss regardless of the cause, providing a comprehensive tool for evaluating the severity of tooth wear in clinical practice. Traditionally attributed to mechanical and chemical insults[7], tooth wear is now increasingly linked to systemic conditions such as obesity, diabetes mellitus, and xerostomia, which may contribute to its pathogenesis through behavioural, dietary, and physiological risk factors[8–10].

Obesity is a chronic metabolic disorder affecting over 650 million adults worldwide and is linked to numerous systemic conditions, including cardiovascular disease, type 2 diabetes, and gastroesophageal reflux disease (GERD)[11,12]. Repeated acid exposure to the teeth, characteristic of GERD, makes it a major intrinsic cause of dental erosion[13]. Hampel et al. have shown a strong association between the prevalence of GERD and elevated body mass index (BMI)[14]. While BMI has commonly been used to assess general obesity, there is now an increasing focus on central obesity, defined by excess abdominal fat and measured by waist circumference or waist-hip ratio[15]. Central obesity, being more metabolically active[16], is more strongly associated with reflux and oesophageal dysfunction[17]. This association is likely due to increased abdominal pressure leading to transient relaxation of the lower oesophageal sphincter, thereby exposing the oral cavity to gastric content[18].

Several studies support the connection between obesity and tooth wear. Kamal et al. in their research on American adults found that those in the overweight and obese groups had 1.72 and 1.61 times, respectively, more worn surfaces than those with normal weight[9]. Similarly, Jastaniyah et al. revealed a significantly higher prevalence of dental erosion among Saudi children and adolescents in the overweight/obese group (8.42%) compared to the control group (2.78%)[19]. These findings suggest that obesity may be a potential risk indicator for dental erosion in children and adults. However, obesity-related patterns and their systemic impacts are influenced by regional and ethnic differences[20]. Specifically, the use of BMI and waist circumference cut-offs specific to Asian populations[21] underscores the importance of accounting for regional body composition profiles when examining the links between obesity, systemic health, and oral health outcomes.

Beyond physical effects, both obesity and tooth wear are now increasingly recognised as factors contributing to reduced Oral Health-Related Quality of Life (OHRQoL)[22,23]. Tooth wear may compromise chewing, speech, and appearance, causing emotional distress and social discomfort[24,25]. This combination of functional and psychological challenges can lead to lower self-esteem and overall dissatisfaction with oral health[25]. Mehta et al. found that individuals having a BEWE score of ≥ 14 reported substantially higher impacts on daily life[26]. Given the bidirectional relationship between obesity and tooth wear, and their combined effect on OHRQoL, comprehensive clinical attention is needed to address these interconnected issues and improve overall well-being of the population.

Previous studies have linked obesity with tooth wear, but most have relied mainly on BMI, examined non-South-Asian populations, or focused on younger age groups.[27] Evidence therefore remains limited on the concurrent assessment of general and central adiposity in relation to tooth-wear severity among South Asian adults. This distinction is important because BMI reflects overall adiposity, whereas waist circumference captures abdominal fat distribution. Accordingly, the present study used Asian-specific BMI action points and sex-specific waist-circumference cut-offs to provide a population-relevant classification of general and central adiposity, rather than tooth-wear-specific diagnostic thresholds[27,28].

This report is part of a broader research study evaluating the frequency and severity of tooth wear, as well as its association with various risk factors. The present analysis explicitly examines the association between anthropometric measures of general and central obesity, namely, BMI and waist circumference (WC) with BEWE-based tooth-wear categories among Pakistani adults. Additionally, this study examines the relationship between tooth wear severity and participants' OHRQoL, as measured by OHIP-14 scores.

Methods

This cross-sectional study received ethical approval by the institutional Ethical Review Committee (AKU-ERC) under reference number 2023-6721-26979 and was performed in accordance with relevant ethical guidelines and regulations, including adherence to the Declaration of Helsinki for research involving human participants.

An anticipated population proportion of 0.40 affected individuals was used, as reported in the reference study by Awad et al. [29] to estimate a sample size of 396 participants, as calculated by the World Health Organisation (WHO) sample size calculator, with a 95% confidence interval and an absolute precision requirement of ≤0.05. After inflating the sample size by 10% to compensate for errors, the sample size increased to 406 participants.

Screening and eligibility

The investigators employed a non-probability consecutive sampling technique for this single-centre study. Adults aged 18 to 65 years, visiting the dental clinics of a tertiary care hospital for a dental consultation, were screened. Eligibility required all six sextants to be recordable for clinical examination, where each sextant contained two or more teeth with less than 50% of their surface area covered by restorations. Patients exhibiting any dental anomalies, or suffering from dental or maxillofacial trauma, having hearing or speech disorders, suffering from neurological conditions, depression, or undergoing orthodontic treatment, radiotherapy, or chemotherapy were excluded. Written informed consent was obtained from all participants.

Study assessment

A clinical examination was conducted on each participant under the dental unit light using a mouth mirror and WHO probe, with gauze used to dry the tooth surfaces to facilitate accurate assessment of tooth wear. Tooth wear was assessed according to the guidelines of the Basic Erosive Wear Examination (BEWE) index[5] by a single trained and calibrated examiner. BEWE score of '0' indicates no signs of tooth wear; '1' depicts initial loss of surface texture, '2' denotes distinct surface defect with <50% of the surface area involved in wear; and '3' indicates ≥50% of the hard tissue loss of surface area. A cumulative BEWE score (sum of scores from all six sextants) ranging from 0 to 18 was calculated and used to categorise participants into low (≤6), medium (7–13), and high-risk (≥14) groups, consistent with thresholds derived from the original BEWE framework proposed by Bartlett et al., and commonly applied in epidemiological studies for simplified risk stratification and analysis[30]. To assess intra-examiner reliability, 40 participants (10% of the sample) were reassessed at recall appointments scheduled within three weeks of initial examination, during their ongoing dental treatment. Intra-rater agreement was calculated using Cohen's Kappa.

Data collection

Data were collected over a period of one year, from January 4, 2023, to January 16, 2024, using REDCap Software (Research Electronic Data Capture; version 13.7.1, developed by Vanderbilt University). Demographic data, including age and sex, weight (in kilograms), and height (in meters), were recorded. Readings of weight and height were used to calculate BMI using the formula: weight (in kilograms) / height² (in meters)[31]. Waist

circumference (WC) was measured at the level of the iliac crest using a non-stretchable measuring tape, with the participant standing upright, the tape held parallel to the floor, and measurements recorded at the end of normal expiration. An average of two readings was taken for each participant. This method was selected due to its standardization and widespread use in clinical and epidemiological studies, although alternative anatomical landmarks have been described in the literature. While waist-to-hip ratio is also considered a complementary measure of central obesity, WC was preferred in this study due to its simplicity, clinical applicability, and strong association with metabolic risk[15]. Excessive risk was defined using WC thresholds of 35.4 inches (≥90 cm) for males and 31.5 inches (≥80 cm) for females[32,33]. All participants completed the OHIP-14 questionnaire, available in both the original English version[34] and a previously validated Urdu translation[35]. The questionnaire consists of 14 items assessing oral health-related quality of life (OHRQoL), with responses recorded on a five-point Likert scale (0 = Never, 1 = Hardly ever, 2 = Occasionally, 3 = Fairly often, 4 = Very often), resulting in a total possible score ranging from 0 to 56. Data on participants' self-reported frequency of exposure to dietary acids, episodes of GERD and related symptoms, and parafunctional habits were collected using a structured interview-based questionnaire designed to assess the frequency of exposure to various tooth wear risk factors.[29,36].

Statistical analysis

Data were analysed using IBM SPSS Statistics for Windows, Version 23.0; IBM Corp., Armonk, NY, USA. Kolmogorov-Smirnov test was used to assess the normality of data for cumulative BEWE scores, WC, weight, BMI, and OHIP-14 scores. Mann-Whitney U test was used to detect statistical significance in differences of these variables among males and females, while Kruskal-Wallis Test was used to detect differences among patients in the tooth wear risk groups. Statistical significance in the differences of frequencies (number and percentages) was analysed for varying categories of BMI and WC among population groups using Chi-square or Fisher's exact test. Multinominal logistic regression analysis was used to derive relative risk ratio using (Stata Software version 18.0) for each unit increase in WC, weight, BMI, and OHIP-14 sum scores, comparing participants in the medium- and high-risk tooth wear groups to those in the low-risk group, as classified by cumulative BEWE scores keeping the confidence interval at 95% and a statistical significance level of ≤0.05.

Results

Of the 406 participants, 263 (64.8%) were males and 143 (35.2%) were females. Signs of tooth surface loss were observed in 99% of the population and were categorised into low, medium, and high-risk levels for tooth wear progression based on cumulative BEWE scores. Eighty participants (19.7%) were identified as being at low risk for tooth wear progression, 233 (57.4%) were at medium risk, and 93 (22.9%) were at high risk. Table 1 presents summary statistics for cumulative BEWE scores, waist circumference, BMI, and OHIP-14 scores of the study population. Intra-rater reliability for BEWE scoring yielded a Cohen's Kappa value of 0.963 ($p < 0.001$), with a percentage agreement of 97.5%.

Table 1 Study population's cumulative BEWE scores, waist circumference, BMI and OHIP-14 scores, with comparisons between male and female participants.

BEWE Cumulative scores	Total	Males	Females	*p*-value
	10.5 (5.25)	12 (6)	9 (5)	<0.001*
Waist Circumference (in inches)	37 (6)	37 (6)	36 (7)	0.089
BMI (Kg/m^2)	26.2 (5.93)	26.7 (5.4)	25.1 (6.6)	0.713
OHIP-14 Sum Score	5 (10.25)	5 (9)	5 (11)	0.317

Median and Interquartile range (IQR) reported for the variables, p-values for differences in variables between males and females reported from Mann-Whitney U test, BEWE; Basic Erosive Wear Examination, BMI; Body Mass Index, OHIP-14; Oral Health Impact Profile, Kg; kilogram, m; meters, *; statistically significant p-value (≤0.05).

Age distribution of the study population was categorised into four groups: 18–30 years, 31–40 years, 41–50 years, and 51–65 years. Two hundred twenty-nine participants (56.4%) fell within the ≤40 years category, with 143 (62.4%) classified as having a medium risk for tooth wear progression (Fig. 1). Table 2 further illustrates the distribution of participants across tooth wear risk categories by age group, showing a statistically significant association ($p < 0.001$).

Figure 2 illustrates the distribution trend of participants at risk for tooth wear progression across different BMI categories. A statistically significant difference was observed in cumulative BEWE scores across the BMI categories ($p < 0.001$, Kruskal–Wallis test), Table 2. Mean rank of BEWE scores increased consistently with higher BMI levels, from 105.95 in the underweight group, to 169.68 in the 'increasing but acceptable risk' group, 213.43 in the 'increased risk' group, and 223.70 in the 'higher high-risk' BMI category, indicating greater tooth wear severity among participants with higher BMI. Post-hoc pairwise Mann–Whitney U tests with Bonferroni correction (adjusted $\alpha = 0.0083$) confirmed significantly higher BEWE scores in the 'increased risk' and 'higher high-risk' BMI groups compared to the underweight group (both $p < 0.001$).

Excessive waist circumference was observed in 174 (66.2%) males (≥35.4 inches or 90 cm) and 110 (76.9%) females (≥31.5 inches or 80 cm), with marked variation across tooth wear risk groups[37]. One hundred fifty-one (86.8%) males and 95 (86.4%) females in the high-risk category for WC also had cumulative BEWE scores ≥7, as shown in Fig. 3. Participants with medium or high risk of tooth wear progression had significantly greater waist circumference values compared to the low-risk group ($p < 0.001$). A significant difference was also observed between the medium- and high-risk groups ($p = 0.003$), Table 2.

A slight upward trend in OHIP-14 scores was observed among study participants across low to high tooth wear risk groups (Fig. 4), although this difference was not statistically significant ($p = 0.956$). No significant differences in OHIP-14 domain scores were found between males and females across the tooth wear risk groups (Table 3).

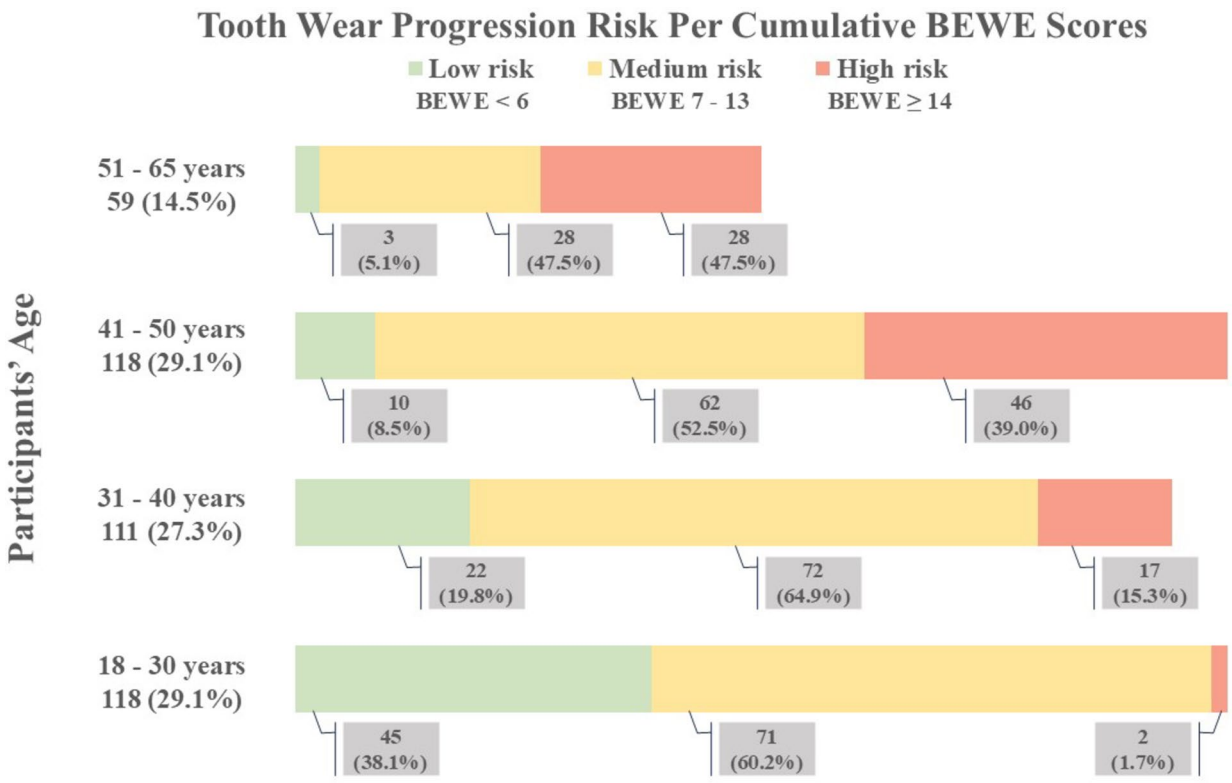


Fig. 1 Age distribution of participants by tooth wear risk category (low, medium, high), based on cumulative Basic Erosive Wear Examination (BEWE) scores, presented as number and percentage.

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Table 2 Distribution of participants in BEWE risk levels, across age groups ≤ 40 and > 40 years, BMI categories, and lower versus excessive risk with WC for males and females.

Risk Factor	Total population	Tooth wear progression risk based on cumulative BEWE scores			*p*-value
		Low risk (BEWE < 6)	Medium risk (BEWE 7–13)	High risk (BEWE ≥ 14)	
Participants age					< 0.001[a]
≤ 40 years old population	229 (56.4%)	67 (29.3%)	143 (62.4%)	19 (8.3%)	
> 40 years old	177 (43.6%)	13 (7.3%)	90 (50.8%)	74 (41.8%)	
BMI					
Underweight (< 18.5 Kg/m^2)	22 (5.4%)	12 (54.5%)	9 (40.9%)	1 (4.5%)	***<0.001** **0.003**[a] for 'Low' vs. 'Medium-risk' among 'Underweight' vs 'Increased risk' groups **0.036**[a] for 'Low' vs. 'Medium-risk' among 'Increased but acceptable risk' vs 'Increased risk' **0.006**[a] for 'Low' vs. 'High-risk' among 'Underweight' vs 'Increased risk' **0.003**[a] for 'Low' vs. 'High-risk' among 'Underweight' vs 'High-risk'
Increasing but acceptable risk (18.5 to 23 Kg/m^2)	74 (18.2%)	23 (31.1%)	38 (51.4%)	13 (17.6%)	
Increased Risk (23–27.5 Kg/m^2)	157 (38.7%)	20 (12.7%)	102 (65.0%)	35 (22.3%)	
Higher high risk (> 27.5 Kg/m2)	153 (37.7%)	25 (16.3%)	84 (54.9%)	44 (28.8%)	
WC in males					
Lower than the cut-off point	89 (33.8%)	22 (24.7%)	50 (56.2%)	17 (19.1%)	**0.014** (**0.009**[a] for low vs. high risk tooth wear)
Excessive risk (≥ 35.4 inches)	174 (66.2%)	23 (13.2%)	94 (54.0%)	57 (32.8%)	
WC in females					
Lower than the cut-off point	33 (23.1%)	20 (60.6%)	13 (39.4%)	0	***<0.001** **<0.001**[a] for low vs. medium and low vs. high risk tooth wear
Excessive risk (≥ 31.5 inches)	110 (76.9%)	15 (13.6%)	76 (69.1%)	19 (17.3%)	

Frequencies (numbers and percentages) reported with p-values (α) generated from the Chi-square test (or *Fisher's exact test). Post-hoc pairwise comparisons with Bonferroni correction (signified with [a]) were done to detect significant p-values among tooth wear risk groups with age, waist circumference (adjusted α = 0.0166), and BMI categories (adjusted α = 0.0027). vs.; versus, WC; Waist Circumference, BEWE; Basic Erosive Wear Examination, BMI; Body-mass index.

Of the seven OHIP-14 domains, functional limitation was the only one showing a statistically significant difference between the high- and medium-risk groups analysed with post-hoc pairwise comparisons using the Mann–Whitney U test. A greater number of participants in the high tooth wear risk group reported feeling that their sense of taste has worsened compared to those in the medium-risk group (Mean Rank: medium-risk = 157.51 vs. high-risk = 178.5; $p = 0.015$). Participants reporting frequent GERD symptoms (≥ 4 times per week) experienced significantly greater changes in their sense of taste compared to both those with less frequent GERD episodes (1–3 times per week; mean rank: 70.59 vs. 54.47; $p = 0.006$) and those who rarely or never experienced GERD (mean rank: 205.48 vs. 155.45; $p < 0.001$).

Multinomial logistic regression analysis revealed that higher waist circumference, weight and BMI were all significantly associated with increased relative risk ratio of being in the medium- or high-risk tooth wear groups compared to the low-risk group (Table 4), as the relative risk ratio for these variables were all greater than 1, indicating a positive association with tooth wear risk.

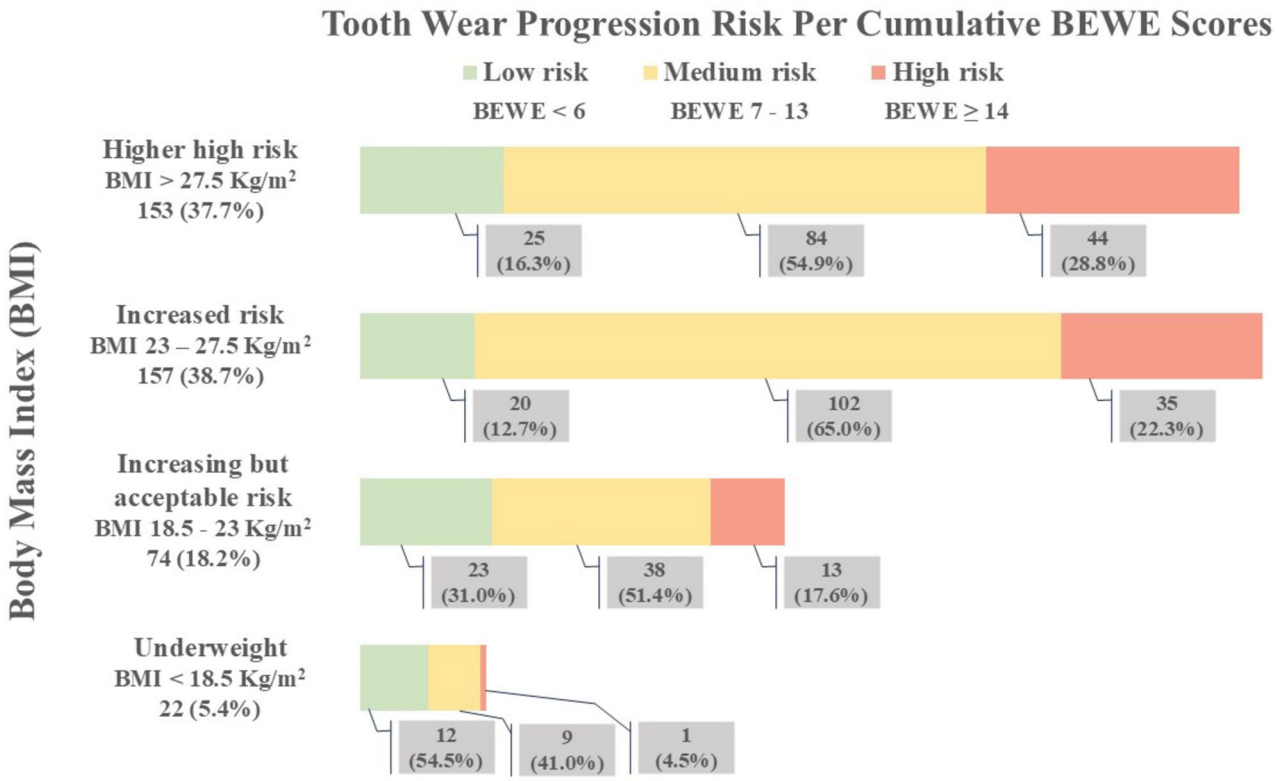


Fig. 2 Distribution of proportions (numbers and percentages) of population with low, medium and high risk for tooth wear progression based on cumulative Basic Erosive Wear Examination (BEWE) scores across different Body-Mass Index (BMI) categories.

Discussion

This study aimed to explore the relationship between anthropometric measures, specifically BMI and WC, and the risk of tooth wear progression, while also evaluating the impact of tooth wear on OHRQoL using OHIP-14 scores. The statistically significant association between increasing age and higher tooth wear risk observed in this study is consistent with previous studies highlighting the cumulative effect of ageing on tooth surface loss over time[1]. However, it is noteworthy that a substantial proportion of younger individuals ($n = 143$, 62.4% of ≤40 years old population) presented with medium risk. This underscores the importance of early detection and timely preventive strategies to mitigate the progression of tooth wear from an early age[7,38].

BMI and WC serve as established indicators of general and central adiposity, respectively[39]. The combined use of these measures is widely recommended, particularly in Asian populations where central obesity is prevalent even at lower BMI thresholds, to enhance the identification of individuals at increased metabolic and cardiovascular risk[40–42]. In this study, both parameters were used to investigate their potential role in tooth wear, an emerging but underexplored association.

An increased WC has been reported to have an adjusted relative risk ratio of 1.264 (95% CI: 1.014–1.582) for the severity of GERD[43]. Abdominal obesity is thought to increase intragastric pressure, alter the gastroesophageal pressure gradient, promote transient lower oesophageal sphincter relaxations (TLOSR), and enhance oesophageal acid exposure—all of which can increase the likelihood of acid reflux episodes reaching the oral cavity[44]. Given the established link between abdominal obesity and GERD[44–46], and the role of GERD as a risk factor for erosive tooth wear[7,10,13], our findings underscore the associated relationship between central adiposity and tooth wear progression.

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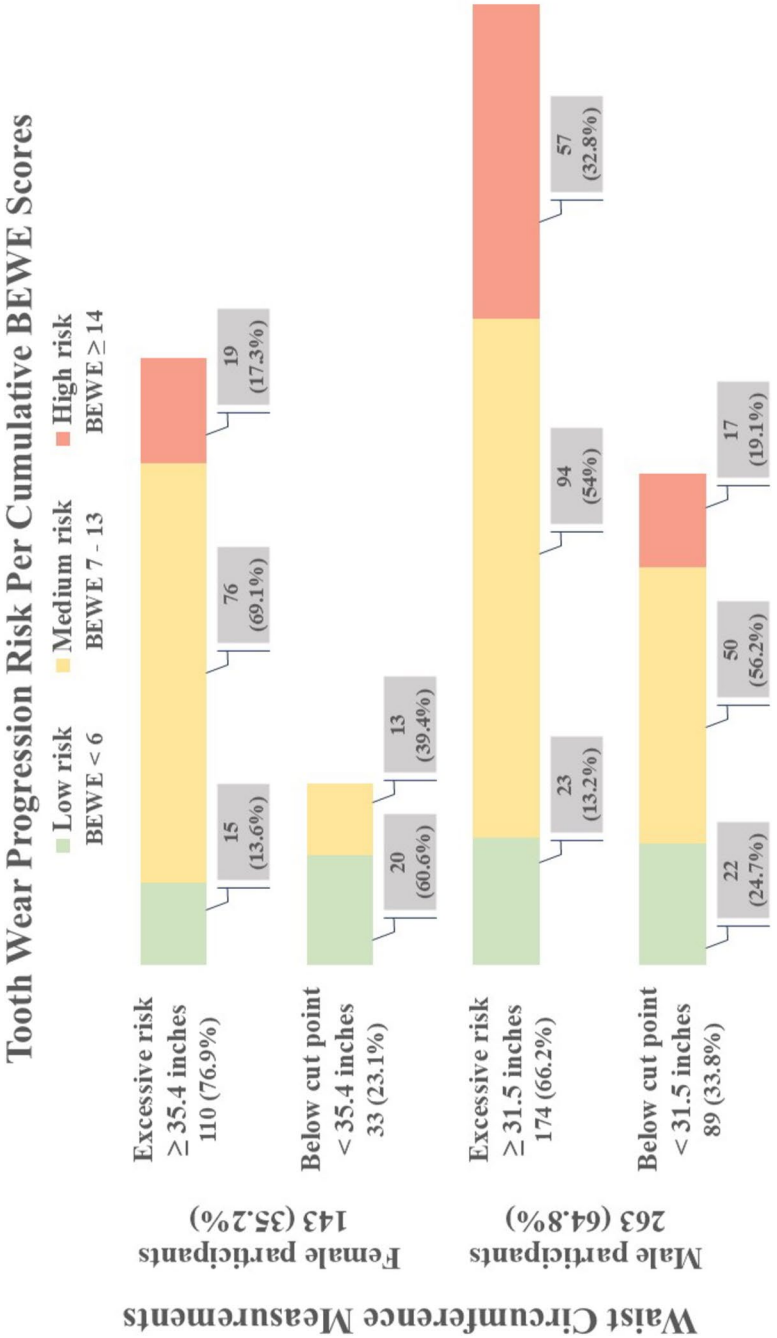


Fig. 3 Waist circumference measurement distribution across proportions (numbers and percentages) of population with low, medium and high risk for tooth wear progression based on cumulative Basic Erosive Wear Examination (BEWE) scores.

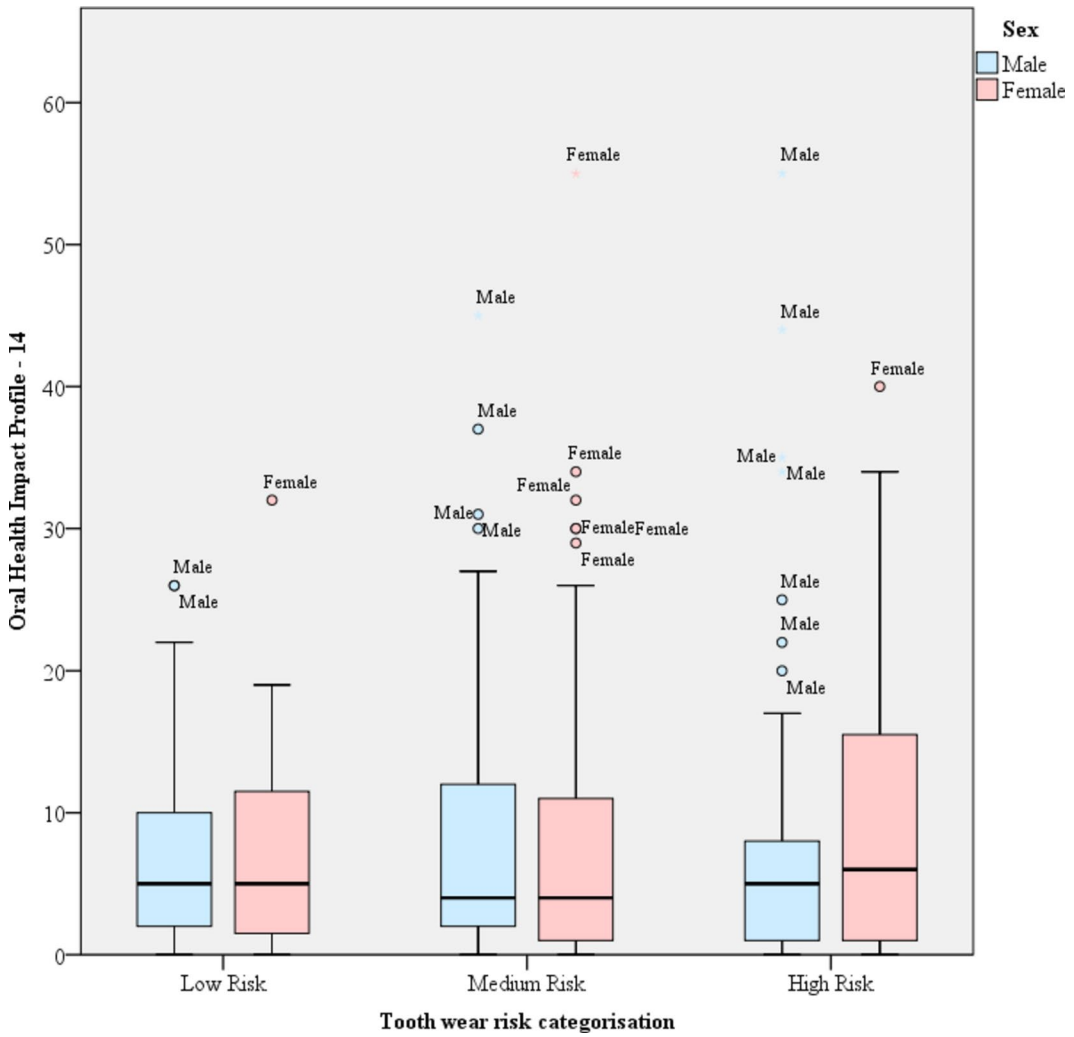


Fig. 4 Distribution of OHIP-14 scores across tooth wear risk categories (low, medium, high) based on cumulative Basic Erosive Wear Examination (BEWE) scores among males and females.

Table 3 Comparison of OHIP-14 domain scores among males and females across low, medium, and high tooth wear risk groups based on cumulative BEWE scores.

OHIP-14 Domains	Tooth Wear Risk Groups						*p*-value	
	Males			Females				
	Low Risk	Medium Risk	High Risk	Low Risk	Medium Risk	High Risk	*Males vs. Females	†Low vs. Medium vs. High Tooth Wear Risk
Functional limitation	0 (1)	0 (1)	0 (2)	0 (1)	0 (1)	0 (2)	0.143	0.099
Physical pain	2 (3)	2 (3)	1 (3)	1 (4)	1 (3)	2 (5)	0.373	0.818
Psychological discomfort	2 (3)	1 (4)	1 (3)	1 (4)	1 (4)	1 (5)	0.622	0.988
Physical disability	0 (2)	0 (1)	0 (1)	0 (1)	0 (1)	0 (4)	0.44	0.827
Psychological disability	0 (2)	0 (2)	0 (2)	0 (2)	0 (2)	1 (3)	0.476	0.637
Social disability	0 (2)	0 (1)	0 (0)	0 (0)	0 (1)	0 (2)	0.932	0.998
Handicap	0 (2)	0 (1)	0 (0)	0 (0)	0 (1)	0 (2)	0.991	0.574

Medians and interquartile ranges (IQR) reported with p-values generated from the *Mann Whitney U test and †Kruskal Wallis test. OHIP-14; Oral health-related impact profile, vs.; versus, BEWE; Basic Erosive Wear Examination, BMI; Body-mass index.

Table 4 Results of multinomial logistic regression analysis reported with relative risk ratio of medium and high risk for tooth wear progression.

Risk Factor	Medium Risk			High Risk		
	Relative Risk Ratio	95% CI	*p*-value	Relative Risk Ratio	95% CI	*p*-value
Waist Circumference	1.142	1.076, 1.212	<0.001*	1.225	1.143, 1.312	<0.001*
Weight	1.029	1.011, 1.048	0.002*	1.044	1.023, 1.066	<0.001*
BMI	1.115	1.052, 1.183	<0.001*	1.156	1.081, 1.236	<0.001*
OHIP-14 Sum Score	1.007	0.978, 1.037	0.64	1.014	0.981, 1.048	0.416

CI; Confidence Interval. BMI; Body Mass Index, OHIP-14; Oral Health Impact Profile, *; statistically significant p-value (≤0.05).

To ensure population-specific relevance, BMI and WC classifications used in this study followed World Health Organisation recommendations tailored for Asian populations[21,37]. While region-specific anthropometric thresholds for the Pakistani population are still lacking, these Asian-specific criteria offer a suitable interim benchmark, given the shared ethnic and metabolic risk profiles[21,47]. Applying these thresholds enabled the identification of statistically meaningful relationships between anthropometric measures and tooth wear risk. However, nationally representative studies are warranted to refine these parameters further.

The association between obesity and tooth wear has previously been linked, in part, to dietary behaviours, particularly the consumption of sugar-sweetened acidic beverages[9]. Our results build on this evidence, showing that participants with higher BMI and WC values were significantly more likely to present with severe tooth wear (Table 4). Specifically, 22.3% of individuals in the 'increased risk' BMI category and 28.8% in the 'higher high risk' category exhibited a cumulative BEWE score of 14 or higher, a threshold indicating advanced tooth wear (Fig. 2). These findings further support the emerging understanding that both general and central obesity are associated with a greater risk of non-carious tooth surface loss.

Comparison with local studies reinforces the generalizability of our WC findings. The median WC in our cohort was 37 inches (IQR 6) (Table 1), comparable to a recent adult survey in Pakistan reporting a mean WC of 36.7±3.5 inches[48]. Another community-based study from Karachi reported elevated waist circumference (≥90 cm in men and ≥80 cm in women) in 73.9% (95% CI: 71.1−76.5) of participants with a mean age of 42.66±12.18 years[49], illustrating a high burden of central adiposity. In our cohort, 32.8% of males and 17.3% of females in the 'excessive risk' category for WC exhibited BEWE scores of 14 or higher. This suggests a meaningful overlap between central obesity and severe tooth wear, given the rising prevalence of central obesity in urban South Asian populations due to sedentary lifestyles and dietary shifts[50,51]. These findings emphasise the need for integrated oral health screening and counselling in systemic disease prevention frameworks.

An additional observation from this study was the statistically significant association between severe tooth wear and both self-reported taste changes and frequent GERD symptoms. This aligns with recent regional data indicating that GERD symptoms are highly prevalent in Pakistani adults, where up to 68% report mild reflux episodes[52], and 60–66% report hyperacidity and bitter regurgitation[53]. While no direct studies have assessed taste perception in patients with tooth wear, it is plausible that chronic acid exposure from GERD could alter oral mucosal integrity, salivary pH, and taste receptor function, thereby contributing to dysgeusia[13,54,55]. Our findings support this hypothesis and call for further exploration of sensory implications in patients with acid-induced tooth wear.

Although a trend of increasing OHIP-14 scores with higher BEWE risk categories was observed (Figs. 4 and 5), the differences did not reach statistical significance. This suggests that, within this relatively healthy adult cohort, tooth wear had a limited perceived impact on daily functioning or psychosocial well-being. This is in line with previous findings indicating that mild to moderate tooth wear does not substantially affect OHRQoL unless it is functionally or aesthetically debilitating[56].

The non-probability, single-centre sampling limits the external validity and generalizability of the results. Additionally, the cross-sectional design restricts our ability to establish causal relationships between obesity and tooth wear. The reliance on self-reported data for key exposure variables, such as GERD symptoms and dietary

acid exposure, introduces potential recall and reporting bias. Furthermore, we did not include objective validation for GERD symptoms or dietary behaviours, which could enhance the robustness of our findings. Lastly, while the study offers valuable insights, the limited novelty of the study and its focus on a specific population means that further research, ideally with more diverse populations and objective assessments, is required to confirm and expand upon these results.

Conclusion

In this single-centre cross-sectional study of Pakistani adults, higher BMI and waist circumference were associated with higher tooth-wear categories, suggesting that both general and central adiposity may be relevant to tooth-wear severity. However, the study design does not establish temporality, causality, or progression, and the study findings may be affected by self-reported GERD symptoms and dietary exposures. Overall OHRQoL did not differ significantly across tooth-wear categories. Multicentre longitudinal studies including objective assessment of reflux and dietary exposures are required to determine whether general or central adiposity predicts the development or progression of tooth wear.

Author contributions F.S did Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Visualisation and Writing – original draft M.K did Conceptualisation, Methodology, Project administration, Software and SupervisionR.S did Writing – original draft and Writing – review and editingT.K did Conceptualisation, Methodology, Project administration, Supervision and Writing – review and editingRubbeya ShahidWriting – original draftWriting – review and editing.

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Data availability The datasets generated from the present study will be freely available to researchers who wish to use them, upon request, from the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

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